

STAT lecture 3

Continuous random variables and probability distributions

Disclaimer:

Please note that these slides are only a brief overview of the topic.
Many more details will be explained on the blackboard and with
MATLAB examples

Continuous random variable

Stochastic experiment

A random variable, X , defines a real number for each of the outcomes in sample space, S , i.e $X: S \rightarrow I \subseteq \mathbb{R}$

e.g outcome a_1 : $X = x_1$

Continuous random variable: the outcomes are not countable (interval subset of $\mathbb{R}$)

Probability density function, pdf

$f_X(x) \stackrel{\text{def}}{=} \frac{P(X \in [x, x+dx])}{dx}$ gives probability densities (likelihoods), not true probabilities

$$f_X(x) \geq 0$$

$$\int_{x \in S} f_X(x) dx = 1$$

$P(a \leq X \leq b) = \int_a^b f_X(x) dx$ pdf must be integrated over an interval to give true probability

Cumulative ("summed from left") probability density function, cdf

$$F_X(x) \stackrel{\text{def}}{=} P(X \leq x) = \int_{-\infty}^x f_X(u) du$$

$$0 \leq F_X(x) \leq 1$$

$$\lim_{x \rightarrow -\infty} F_X(x) = 0$$

$$\lim_{x \rightarrow \infty} F_X(x) = 1$$

$$f_X(x) = \frac{dF_X(x)}{dx}$$

$$P(a \leq X \leq b) = \int_a^b f_X(x) dx = F_X(b) - F_X(a)$$

Mean value of X

$$\mu_X = E[X] \stackrel{\text{def}}{=} \int_{x \in S} x f_X(x) dx$$

Variance of X

$$\sigma_X^2 = V[X] \stackrel{\text{def}}{=} E[(X - \mu_X)^2] = E[X^2] - \mu_X^2 = \int_{x \in S} x^2 f_X(x) dx - \mu_X^2 \geq 0$$

Continuous uniform distribution

$$X \sim U(a, b), \quad a, b \in \mathbb{R}$$

$$f_X(x) = \frac{1}{b-a}, \quad a \leq x \leq b$$

$$\mu_X = \frac{a+b}{2}$$

$$\sigma_X^2 = \frac{(b-a)^2}{12}$$

Normal (Gaussian) distribution

$$X \sim N(\mu, \sigma^2), \quad E[X] = \mu, \quad V[X] = \sigma^2$$

$$f_X(x) = \frac{1}{\sqrt{2\pi}} \frac{1}{\sigma} \exp\left(-\frac{1}{2} \left(\frac{x-\mu}{\sigma}\right)^2\right) \quad x \in \mathbb{R}$$

Standard normal $Z \sim N(0, 1)$

Linear combinations of independent normal distributions gives new normal distribution with summed means and summed variances from linear combination components

Normal distribution approximation to binomial distribution

$$X \sim \text{Bin}(n, p) \quad \text{with} \quad np \gg 1, \quad np(1-p) \gg 1$$
$$Z = \frac{X - np}{\sqrt{np(1-p)}} \sim N(0, 1) \quad (\text{approximately})$$

Normal distribution approximation to Poisson distribution

$$X \sim \text{Poisson}(\lambda)$$
$$Z = \frac{X - \lambda}{\sqrt{\lambda}} \sim N(0, 1) \quad (\text{approximately})$$

Exponential distribution

Waiting time for next arrival in Poisson process

$$T \sim \text{Exp}(\lambda), \quad \lambda \geq 0$$
$$f_T(t) = \lambda e^{-\lambda t}, \quad t \geq 0$$
$$E[T] = \frac{1}{\lambda}, \quad V[T] = \frac{1}{\lambda^2}$$

Central Limit Theorem

Why is the Normal distribution "*normal*"

$X_1, X_2, \dots, X_n$ independent and identically distributed with $E[X_i] = \mu, V[X_i] = \sigma^2$

Then whatever distribution X_i might have

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i \sim N\left(\mu, \frac{\sigma^2}{n}\right) \quad \text{for } n \rightarrow \infty$$

and equivalently

$$Z = \frac{\bar{X} - \mu}{\sqrt{\frac{\sigma^2}{n}}} \sim N(0, 1) \quad \text{for } n \rightarrow \infty$$